EXATHON INDIA 2026 · TRACK 3

VoltProof

Traceable evidence of power cuts and poor supply, from minute-by-minute voltage records, analysed inside Exasol.

> Households and small shops on unreliable electricity can't show when, and for how long, their power failed, because when they complain, their own memory is usually all they have to set against the utility's records. If this worked, they could take a dated, minute-by-minute record of every outage and voltage problem, with its uncertainty stated, to their utility or grievance forum.

Data: Prayas (Energy Group), Electricity Supply Monitoring Initiative (ESMI), Harvard Dataverse doi:10.7910/DVN/CLLZZM (CC0 1.0).

github.com/stredevirad/voltproof-exathon-2026

THE PROBLEM

A consumer's memory against the utility's own reports

- Daily power cuts and voltage too low or too high to use are common in parts of India.
- Complaints rest on memory; there is rarely an independent, dated record.
- Prayas's ESMI monitors recorded supply every minute at homes and shops, and published it, but as raw 63-column hourly files with gaps, duplicates and defects.

Illustrative A shopkeeper in a small Uttar Pradesh town loses business to evening cuts but cannot show when, how long, or whether neighbours lost supply too.

Rural Uttar Pradesh, 2016–2018

566.3 min/day

measured outage (lower bound), 82 rural monitors, comparable months, VoltProof

Prayas's own article reports "in excess of 4 hours (>240 minutes) a day" for rural areas; our measurement is higher (see validation).

Sources: docs/HEADLINE_FIGURES.md; Prayas, "ESMI in Uttar Pradesh", 7 Aug 2019.

WHO IT HELPS

Evidence anyone can check, not a verdict

Consumers and consumer groups

A dated PDF per location-month: every interruption, its certainty and bounds, and a pointer to the grievance route.

Researchers and civil society

Reproducible, versioned SQL over the raw files; data defects found and documented instead of silently dropped.

Grievance forums and regulators

Every number traces to a source file line and an Exasol statement id, so a reviewer can verify it rather than trust it.

VoltProof is an evidence-analysis aid. It does not state what anyone is owed, does not name a cause, and is not legal advice.

DEMO MOMENT · BAHRAICH, 25 JULY 2018, 18:27

17 monitors lost supply within 2 minutes

16 other monitors lost supply within 2 minutes

Potential shared supply event Bahraich | 2018-07-25 18:27: 136 coincident pairs in Bahraich district. 21 monitors in the district had interruptions that month.

Monitor	Start	Offset	Measured
Huzurpur-Bahraich	**2018-07-25 18:27**	**+0 min**	**1 h 07 min**
Mihinpurwa-Bahraich	2018-07-25 18:27	+0 min	37 min
Agarwal Mohalla Jarwal Kasba	2018-07-25 18:29	+2 min	32 min
Bahadurpur-Bahraich	2018-07-25 18:29	+2 min	1 h 10 min
Bardaha-Bahraich	2018-07-25 18:29	+2 min	49 min

- One evening outage at Huzurpur-Bahraich: exact start and end, certainty of each edge, the raw source lines.
- 16 other Bahraich monitors started an interruption between 18:27 and 18:29: a **potential shared supply event**.
- Bahraich, July 2018: 3,945 coincident pairs observed vs 1,827.5 expected from 20 time-shifted variants; none reached the observed count.
- Chosen by a fixed, documented rule, not hand-picked.

Coincidence is not cause.

HOW IT WORKS

The whole analysis is versioned SQL running inside Exasol

Verify
MD5 against Dataverse

→

IMPORT
raw rows keep file id and line number

→

Unpivot
63 columns → minutes; per-file date formats; quarantine

→

Runs
voltage classes; gaps-and-islands with one window function

→

Events
certainty and bounds; daily, monthly, coverage

→

Coincidence
pairs across monitors; chance baseline

- **Missing data is never an outage.** Gaps stay NO_DATA; outages next to gaps get a measured lower bound (the headline) and an upper bound.
- A month is comparable only with > 90% of minutes read (Prayas's rule).
- FastAPI + React app and a one-click PDF report; every number traced to the source line and the Exasol statement id in the audit log.
- Thresholds are Prayas's published ones: outage ≤ 130 V, low 131–204 V, high above 270 V.

WHY EXASOL · WARM MEDIANS, 15.6 GB LAPTOP

102.8 million minutes, full build in 210.4 s

30.3 s
native IMPORT of 7,367,581 raw rows (11 files)

11.0 s
in-database unpivot to 102,927,720 minute rows

35.6 s
gaps-and-islands over 102.8 million readings (one window function)

SCALE FIX (RECORDED IN EXASOL'S AUDIT LOG)	BEFORE	AFTER
Run detection: LAG + running SUM → one ROW_NUMBER window	665 s, 7.4 GiB temp	35.6 s
Duplicates: minute-level GROUP BY → hour-level first	1,892 s, then a cancelled step	13.5 s
Chance baseline: band join → bucketed equality joins	348 s (sample only)	3.9 s (full data)

In all 6 benchmark builds, every statement's session and statement id matched EXA_DBA_AUDIT_SQL by exact SQL text (0 different, 0 missing); the same audit statistics produce these benchmarks. Sources: docs/BENCHMARKS.md, docs/WHY_EXASOL.md.

CORRECTNESS EVIDENCE

The SQL must equal an independent implementation, exactly

- **18 hand-written fixtures** (rules R-1 to R-21), expected values derived by hand.
- **Independent plain-Python oracle**: identical to the SQL on every fixture, on **40 random Hypothesis batches**, and on real Uttar Pradesh slices (4,930 events; 20,127 coincident pairs).
- **Harness sensitivity**: a single wrong value is caught.
- **Report consistency**: every number in the PDF equals an independent SQL query and the rendered page.
- **Provenance**: every statement of every build matched to Exasol's audit log by exact SQL text (0 different, 0 missing in all 6 benchmark builds).
- **End-to-end** browser tests of the guided demo, plus a scan for blocked wording on every page and report.

docs/VALIDATION.md §1 · docs/TEST_REPORT.md

VALIDATION AGAINST PRAYAS'S PUBLISHED FIGURES

Agrees at the level of single monitors; headline averages differ

PRAYAS FIGURE	PRAYAS	VOLTPROOF	
Single-monitor profiles, 2017–2018 (4 monitors, 34 monitor-months)	hours/day per class	largest difference 0.2 h/day	✓ agrees
Evening outage (Figure 9), min: megacity / other municipal / rural	12 / 53 / 144	11.3 / 47.8 / 147.3	✓ agrees
Outage by time of day (Figure 8), summed, min/day	≈ 56 / ≈ 155 / ≈ 523	56 / 174 / 555	✓ agrees
Daily outage (Figure 2), min/day	21 / 87 / 260	55.6 / 174.4 / 555.2 2016–2018 only: 56.3 / 189.5 / 566.3	≠ higher

Our daily-outage averages are higher than Prayas's Figure 2; Prayas's Figure 2 and Figures 8–9 imply different averages, and the article does not state the averaging used for Figure 2. No parameter was tuned to close the gap.

docs/VALIDATION.md §3. Primary definition, fixed before comparing: comparable months, 2016-01 to 2019-06, pooled. Headline figures elsewhere use 2016–2018 (the 2019 file defect, D-031); that variant was already reported in §3.3.

DATA DEFECTS FOUND IN THE PUBLISHED FILES

Found, documented and handled, never silently dropped

2019 Jan–June: all-0 V hours omitted ! lower bounds

In 2016–2018, 11–17% of read location-hours are entirely 0 V; in 2019, 4.3%. 82% of 2019 gaps have 0 V just before and after. Found by comparing with Prayas's own single-monitor charts.

Handled: gaps stay missing data; 2019 figures are labelled lower bounds in the app and reports; headlines use 2016–2018.

2019 July–Dec: Hour column lost

690,915 of 695,974 rows say hour 0, so minute timestamps cannot be determined. Loaded, verified, excluded with that reason.

2019 formats

Dates M/D/YYYY (earlier DD-MM-YYYY), detected per file; names with an added " [Offline]", matched only when unambiguous.

What VoltProof does not claim

- A plug-in monitor is not the utility's meter; an outage may also come from a tripped breaker, wiring or the monitor.
- ESMI data is historical (2014–2019): it describes supply then, not today.
- A potential shared supply event is a coincidence in time, not proof of a common cause.
- No amount owed and no breach is stated. Not legal advice.

Grievance route the report points to

In Uttar Pradesh, the Commission's regulations under Section 42(5)–(8) of the Electricity Act, 2003 provide a Consumer Grievance Redressal Forum (s. 42(5)) and an Electricity Ombudsman (s. 42(6)).

UPERC CGRF and Ombudsman Regulations, 2003; PIB, Ministry of Power, 30 Dec 2010. The Electricity (Rights of Consumers) Rules, 2020 (PIB, 21 Dec 2020) postdate the data and are not applied to it.

FUTURE SCOPE (NOT CLAIMED TODAY)

The same pipeline, on today's data

Live data

New plug-in monitors or smart-meter interval data through the same versioned SQL steps.

Standards of performance

Each state's rules, encoded from the primary regulation, to show which events may warrant review through the grievance forum.

More states, maps

The ESMI files cover many states; a map needs coordinates the source does not have.

TEAM

Rajagiri School of Engineering & Technology, Kochi

Joshua Herbert Japheth

Rohith Mohandas

Dane John Padinjarekara

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